Muhammad Akbar Khan



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Research Profile

MS Applied Mathematics researcher at NED University specializing in physics-informed neural networks and neural operators for PDE-governed problems, with a focus on level-set interface advection. Author of five research papers: one published in *Machine Learning: Science and Technology*, one under revision at *Scientific Reports*, two under review at *Machine Learning: Science and Technology*, and one preprint. Reviewer for *Machine Learning: Science and Technology*. Seeking a fully-funded PhD position to advance scientific machine learning at the intersection of numerical analysis and deep learning.

Research Interests

Physics-Informed Neural Networks (PINNs) · Neural Operators · Scientific Machine Learning · Data-Driven PDE Modeling · Numerical Analysis · Level-Set Methods for Multiphase Flow · Computational Heat Transfer

Education

NED University of Engineering & Technology Aug 2024 – Dec 2026 (expected)
Master of Science (MS) in Applied Mathematics *Karachi, Pakistan*

- CGPA: **3.44/4.00**
- Thesis (6 credit hours): *Applications of Physics-Informed Neural Networks for Reinitialization of Level-Set Functions in Interface Tracking Problems*
- Thesis defence scheduled for **8 December 2026**; coursework (24 credit hours) completed
- Supervisor: Dr. Fahim Raees (Associate Professor; PhD, TU Delft, 2016)
- Coursework:
  - *Core mathematics:* Differential Equations, Linear Algebra, Advanced Discrete Mathematics
  - *Computational methods & optimization:* Numerical Methods and Applications, Scientific Computing, Operations Research and Optimization
  - *Statistics & machine learning:* Applied Statistics, Statistical Method and Data Analysis, Fuzzy Logic and Neural Networks

International Islamic University Islamabad Feb 2017 – Jan 2021
Bachelor of Science (BS) in Mathematics *Islamabad, Pakistan*

- CGPA: **3.37/4.00**
- **HEC Need-Based Scholarship** (Higher Education Commission, Pakistan): full tuition and stipend for the duration of the BS program.

Publications

[1] M. Akbar Khan and F. Raees. *A Systematic Study of Physics-Informed Neural Networks for the Level-Set Interface Advection.* **Machine Learning: Science and Technology** (IOP Publishing), 2026. Published July 2026. Gold Open Access, CC BY 4.0.
doi:10.1088/2632-2153/ae8b74.

[2] M. Akbar Khan, F. Raees, and U. Fatima. *A Physics-Informed Neural Operator for Thermal Ranking of Low-Cost Wall Materials in Hot-Dry Climates.* **Scientific Reports** (Springer Nature), 2026. Submitted July 2026; major revision decision September 2026.
arXiv:2607.25668.

[3] M. Akbar Khan. *SDF-Aware Weighting: Adaptive Eikonal Regularisation for Three-Dimensional Level-Set Physics-Informed Neural Networks.* **Machine Learning: Science and Technology** (IOP Publishing), 2026. Under review; submitted August 2026 (MLST-105968).
arXiv:2608.08322.

[4] F. Raees and **M. Akbar Khan**. *A Fourier Neural Operator Surrogate for Hygrothermal Ranking of Indigenous Wall Materials in Hot-Dry Climates.* **Machine Learning: Science and Technology** (IOP Publishing), 2026. Under review; submitted September 2026 (MLST-106107).
doi:10.5281/zenodo.23371804.

[5] M. Akbar Khan. *A Data-Free Physics-Informed Neural Operator for Level-Set Interface Advection.* Preprint, 2026.
arXiv:2609.38195.

Research Experience

NED University of Engineering & Technology March 2026 – Present
Research Assistant, Dept. of Mathematics *Karachi, Pakistan*

- Appointed under **Sindh Research Project 321**: mathematical modeling for heat ventilation in rural housing.
- Developing finite-difference solvers and PINN models for heat transfer simulation in indigenous wall assemblies under Dr. Fahim Raees.

Peer Review

Reviewer for *Machine Learning: Science and Technology* (IOP Publishing), 2026: one manuscript reviewed; review verified on Web of Science (ResearcherID: RFT-1150-2026).

Teaching Experience

Cadet College Ormara Aug 2022 – Dec 2025
Lecturer of Mathematics & OI/C FBISE Affairs *Gwadar, Pakistan*

- Delivered mathematics lectures and tutorials at intermediate and higher-secondary level; designed and assessed examinations.
- Managed all Federal Board of Intermediate & Secondary Education (FBISE) academic affairs: exam registrations, compliance with board regulations, and direct correspondence with FBISE.

Excelsior School System May 2021 – May 2022
Mathematics Teacher *Islamabad, Pakistan*

- Planned and delivered mathematics lessons; set and marked examinations following the national curriculum.

Technical Skills

Scientific ML	PyTorch, Physics-Informed Neural Networks, Fourier Neural Operators, Physics-Informed Neural Operators (PINO), Random Fourier Features, L-BFGS / Adam optimization
Programming	Python (NumPy, SciPy, Pandas, Matplotlib, Statsmodels, SQLAlchemy), SQL / PostgreSQL
Numerical Methods	Finite-difference schemes (WENO5 with TVD Runge–Kutta, Crank–Nicolson), classical ODE/IVP solvers, numerical linear algebra
Tools	Git, GitHub, VS Code, PyCharm Pro, Jupyter, LaTeX (MiKTeX), TikZ
Computing	GPU training on NVIDIA Tesla T4, PyTorch 2.10, Python 3.12

Open-Source Project

PINN Level-Set Solver AkbarTheAnalyst/pinn-level-set

PyTorch implementation underlying published MLST paper (doi:10.1088/2632-2153/ae8b74)

- Full PINN pipeline: 8-layer MLP with Tanh activations and Xavier initialization; RFF input embedding; causal weighting.
- Four canonical benchmarks (translation, rotation, reversed vortex, Zalesak disc) with automated ablation infrastructure.

Certifications

- AI for Everyone, DeepLearning.AI (Coursera)
- Google Data Analytics Professional Certificate, Google (Coursera)

Languages

English: Professional working proficiency · **Urdu**: Native

References

Dr. Fahim Raees (MS Supervisor)
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Dr. Ubaida Fatima
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